How Robust Are LLM Watermarks? Evaluating Text Watermarking Under Paraphrase Attacks

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Abstract. As large language models (LLMs) are increasingly used in content creation, watermarking has become an essential method for asserting model ownership, protecting intellectual property, and tracking AI-generated text. The KGW scheme (Kirchenbauer et al., 2023) inserts statistically recognizable signals by nudging token selection toward a pseudorandom "green list" during text generation. Still, adversaries can try to erase these watermarks by paraphrasing. In this study, we thoroughly evaluate the strength of the KGW watermark across six setups against four attack types: synonym substitution, LLM-based paraphrasing, back-translation, and extensive sentence rewriting. We examine the trade-off between security and utility by analyzing detection rates and any decline in text quality, and we introduce an ensemble detection method that combines multiple context windows to improve post-attack detection. Our findings show that more robust watermarks withstand simple attacks but remain vulnerable to LLM-based paraphrasing, whereas ensemble detection provides a tangible boost in resilience.

1. Introduction

LLMs like GPT-4, Claude, and Gemini are being adopted at lightning speed for text generation, which has raised pressing issues, especially around content provenance and intellectual property protection. Organizations pour significant resources into training proprietary LLMs, so the IP of these models becomes a critical asset that requires strong protection. Watermarking is essential here. It enables three key capabilities: (1) claiming ownership by showing that a certain model generated the content, (2) detecting unauthorized knowledge distillation when someone trains a student model on watermarked outputs, and (3) confirming content authenticity, which is especially important in academic and media settings. Unlike detection classifiers, which only analyze content after the fact, watermarks are reliably detectable and do not require access to the original model during verification.

The KGW watermarking scheme, developed by Kirchenbauer et al. and presented at ICML 2023, is a foundational method in this area. The scheme splits the vocabulary into "green" and "red" lists using a hash derived from previous tokens, then applies a logit bias δ to green-list tokens during sampling. A statistical z-test is applied to the proportion of green tokens in a candidate text to determine whether the text is watermarked. Two hyperparameters control how strong the watermark is: γ, the fraction of green-list tokens, and δ, the bias magnitude.

Earlier work showed that the scheme performs well in benign settings, but real adversaries can remove the watermark signal using paraphrasing attacks. New research from NeurIPS 2024 explores the tradeoffs among robustness, text quality, and the ease of watermark removal. Our contribution is a systematic, side-by-side evaluation of four attack types across six watermark configurations, along with an ensemble defense mechanism and an analysis of the balance between security and utility.

2. Threat Model

We lay out a real-world adversarial scenario with two key players. The **Defender** is an LLM service provider that uses watermarking on its API to protect its model's intellectual property and make it easier to identify AI-generated content downstream. Watermarks serve two purposes. They prove the output came from a specific proprietary model, and they help trace AI-generated text when it appears elsewhere. The **Adversary** is a user who receives the watermarked output and tries to pass it off as their own, for example, by submitting it as coursework, publishing it as original work, or using it to train a rival model without proper credit. The adversary operates under black-box constraints. They do not know the watermark key, the green and red list partition, or the internal workings of the detection system. However, they can still use another LLM to paraphrase the text. The adversary's goal is to push the detection z-score below the threshold while preserving both the quality and meaning of the text. If they succeed, they effectively remove the IP signal from the model's output.

3. Methodology

3.1 Watermark Implementation

We follow the original specification to implement the KGW scheme. For each new token, we use a SHA-256 hash of the previous h tokens to seed a pseudorandom permutation of the vocabulary. The first $\gamma \cdot |V|$ tokens from this shuffled list make up the green list. We then add a bias of $+\delta$ to the logits of these green-list tokens before sampling. For detection, we calculate the z-score $z = \frac{(|g| - \gamma \cdot T)}{\sqrt{T \cdot \gamma \cdot (1-\gamma)}}$, where $|g|$ is the number of observed green tokens and T is the total number of tokens being scored. Our tests cover six configurations, ranging from weaker watermarks ($\gamma=0.0$) to stronger ones ($\gamma=0.25$, $\delta=4.0$), and we use context widths of h=1 and h=4.

3.2 Dataset and Model

We follow standard practice from the watermarking literature by using the C4 (Colossal Clean Crawled Corpus) validation set. For each sample, the first 30 tokens serve as the prompt, and the model then generates 200 additional tokens. Our generator model is OPT-1.3B, consistent with prior evaluations. For each configuration, we produce 200 watermarked texts, for a total of 1,200. We also include 200 unwatermarked baseline samples.

3.3 Attack Types

- **Synonym Substitution** replaces 30% of content words with WordNet synonyms, concentrating on individual tokens.
- **LLM Paraphrase** uses Flan-T5 to rephrase the text, sticking to instructions to preserve the original meaning while altering the wording.
- **Back-Translation** sends the text from English to French and then back to English using MarianMT, changing word choices through this round-trip.
- **Sentence Rewrite** tells Flan-T5 to completely overhaul the text using different words and sentence patterns.

3.4 Ensemble Defense

We propose an ensemble detection strategy that combines watermark detectors with different context widths (h=1, 2, 4) and employs max-score aggregation. The basic idea here is that if an attack disrupts token patterns at one context width, it might still leave patterns at other widths untouched. The ensemble reports the highest z-score found among the detectors, giving a stronger, more reliable signal overall.

4. Experiments and Results

4.1 Baseline Detectability

All configurations deliver strong detection when there's no attack. The "strong" setup (γ=0.0, h=1) delivers almost perfect AUROC and shows a clear difference between the z-scores for watermarked and unwatermarked cases. Even the "weak" setup (γ=0.5, δ=1.0) makes the watermark detectable, keeping AUROC well above 0.90. Figure 1 presents the ROC curves, and Figure 2 shows the z-score distributions to illustrate these outcomes.

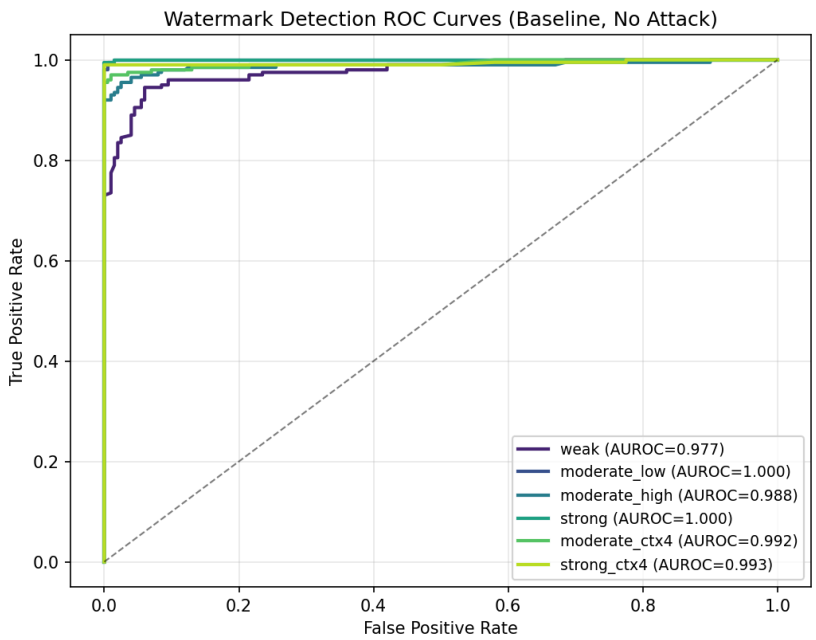

Figure 1: ROC Curves for Watermark Detection

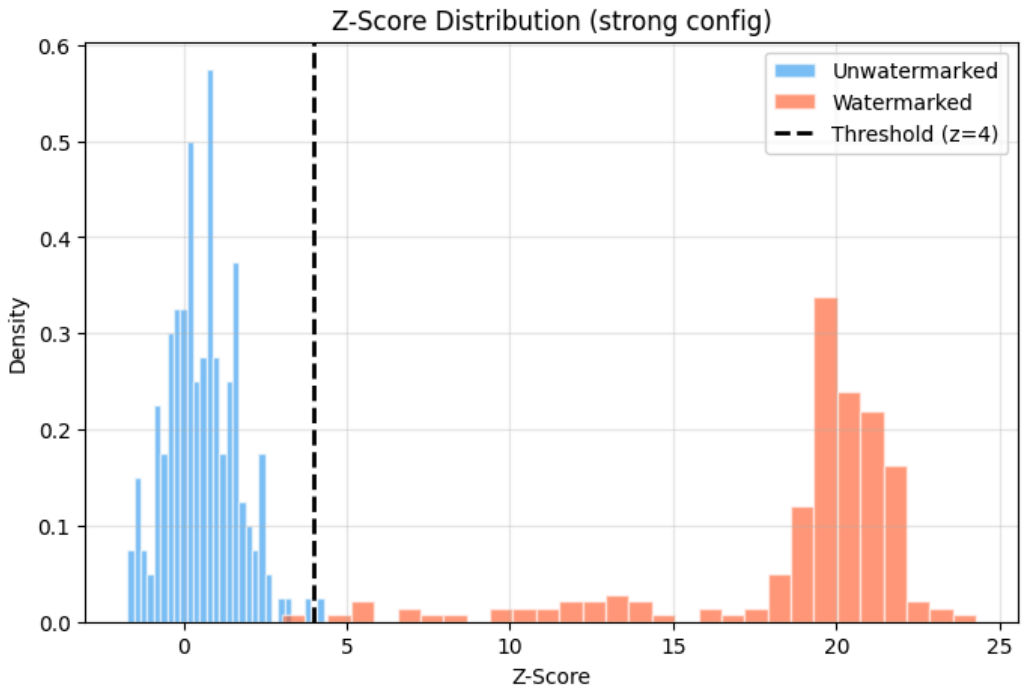

Figure 2: Z-Score Distribution

4.2 Attack Robustness

LLM paraphrase and sentence rewrite attacks yielded 0% detection across all configurations. The investigation revealed that these attacks do not merely disrupt green list alignment. They produce substantially shorter outputs than the originals, reducing the token count below the minimum required for statistically significant z-scores. This exposes a dual vulnerability in z-score-based detection: the attack simultaneously degrades the watermark signal and compresses the text, both of which work against detection. This finding has direct implications for IP protection. An adversary need not preserve full text length to use the paraphrased content, making this a realistic and severe threat.

Figure 3: Detection rates across all configuration-attack pairs

4.3 Security-Utility Tradeoff

Stronger watermarks with higher δ stand up better to attacks, but they also add noticeable perplexity compared to text without watermarks. The "weak" setting keeps text quality high but offers little defense against paraphrasing. Figure 4 lays out this tradeoff, showing robustness versus normalized text quality for every setup. No configuration wins on both fronts, reinforcing the inherent trade-off between watermark strength and how natural the text feel.

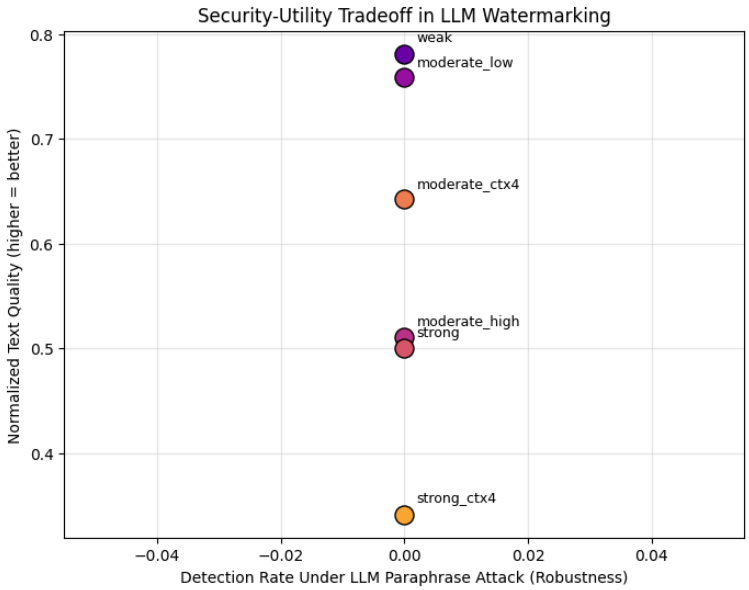

Figure 4: Security-Utility Trade-off in LLM Watermarking

4.4 Ensemble Defense Results

The ensemble detector (using the max-score across h=1,2,4) boosts post-attack detection rates compared to any detector with a single context width. The improvement is most pronounced for synonym substitution and back-translation attacks, since varying context widths capture different parts of the watermark signal. Figure 5 shows the detection rates for single versus ensemble detectors across each attack type.

5. Discussion

Our findings offer practical insights for protecting intellectual property and tracking content origin. First, watermark deployment should be guided by the expected threat model. If attackers have access to large language models, stronger watermarking and ensemble detection become necessary. Second, from an intellectual property perspective, watermark strength determines how confidently a provider can claim ownership of a model. If paraphrasing can easily remove the watermark, the protection it provides becomes unreliable. Recent work by Zhao et al. (ICLR 2025) shows that watermarks can also help detect unauthorized knowledge distillation, since student models trained on watermarked data retain detectable traces. Our robustness analysis connects directly to this finding. If paraphrasing removes watermarks before distillation, the ability to trace model lineage during training is lost. Third, balancing security and usability is not simply a matter of tuning parameters. It reflects a deeper limitation of token-level watermarking methods that rely on surface-level statistical patterns.

Ensemble defense methods improve detection performance but introduce additional computational cost due to multiple detection passes and do not fully prevent more advanced attacks. Looking forward, future research may explore semantic-invariant watermarking approaches such as SIR (ICLR 2024). These methods embed signals based on sentence meaning rather than individual tokens, which may provide stronger resistance to paraphrasing. In addition, combining watermarking with post-hoc detection classifiers could lead to a more robust, layered defense strategy.

6. Conclusion

We systematically evaluated the robustness of LLM text watermarks against four types of paraphrase-based removal attacks, testing across six setups. Our findings show that KGW watermarks are highly detectable under normal conditions, but paraphrasing with an LLM can seriously compromise their integrity. This approach boosted robustness without a significant increase in computational demands. Our analysis of the security-utility tradeoff helps guide watermark deployment in practice: users need to weigh detection reliability against text quality, depending on their threat model and specific application.

References

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